DSA REFERENCE

Binary Search Variants Problem Ladder

A sequenced practice ladder from warm-up to interview stretch.

Search a monotonic decision space by preserving an interval guaranteed to contain the answer.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / BINARY-SEARCH

DSA

Binary Search Variants Problem Ladder

CORE IDEA

Search a monotonic decision space by preserving an interval guaranteed to contain the answer.

Recognition cues

+ sorted data
+ first or last occurrence
+ minimum feasible value
+ monotonic true/false predicate

Implementation sequence

+ choose interval convention
+ write the monotonic predicate
+ bias midpoint when needed
+ state what loop termination means

Invariant

The answer remains inside the chosen closed or half-open interval after every update.

Worked reasoning

For the first feasible capacity, if mid works keep mid and everything larger; otherwise discard through mid.

Complexity

O(log n) predicate calls, multiplied by the cost of one predicate evaluation.

Failure modes

- [ ] mixing interval conventions
- [ ] infinite loops near adjacent bounds
- [ ] searching a non-monotonic predicate
- [ ] overflow in midpoint arithmetic

Practice ladder

Binary Search Variants Problem Ladder: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Binary Search**	State the cue, invariant, edge case, and proof before implementation.
2	**Search in Rotated Sorted Array**	State the cue, invariant, edge case, and proof before implementation.
3	**Find Minimum in Rotated Sorted Array**	State the cue, invariant, edge case, and proof before implementation.
4	**Koko Eating Bananas**	State the cue, invariant, edge case, and proof before implementation.
5	**Median of Two Sorted Arrays**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.